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REVIEW

Sex differences in resistance training: a brief narrative review

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Abstract

Purpose Sex differences in response to resistance training (RT) remain a critical topic in sports science, influencing performance optimization, injury prevention and exercise prescription. Historically, RT research has been male-dominated, creating gaps in understanding female-specific adaptations.

Aims This brief narrative review synthesizes the current literature on sex-based differences in maximal strength, muscle hypertrophy, speed strength, and strength endurance, exploring underlying physiological, anatomical, and neuromuscular factors.

Methods A semi-structured literature search was conducted across sport science databases. Preference was given to studies published between 2020 and 2025, with additional articles identified through screening the references of recent publications. Only peer-reviewed studies on human adults and written in English were included.

Results Males typically exhibit greater absolute strength than females due to higher muscle mass and a larger proportion of type II fibers. However, relative strength gains and hypertrophic responses are largely comparable between sexes. Females demonstrate greater fatigue resistance than males, particularly during isometric contractions. They also tend to exhibit a higher proportion of type I fibers and this provides a partial explanation for their superior endurance capacity.

Conclusions Despite the growing interest in unraveling sex differences in response to RT, the time course of chronic adaptation remains insufficiently studied, especially in females, making it unclear whether both sexes progress at the same rate over time. Given the substantial variability in the responses to RT, this review emphasizes the necessity of individualized training approaches over generalized sex-based prescriptions.

Keywords Exercise · Strength · Hypertrophy · Speed strength · Strength endurance

Abbreviations

1RM	1 Repetition maximum
CON	Concentric
CSA	Cross-sectional area
ECC	Eccentric
FL	Fascicle length
ISO	Isometric
LVP	Load-velocity profile
RT	Resistance training
MVC	Maximal voluntary contractions
RFD	Rate of force development
MT	Muscle thickness
VBT	Velocity-based training

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Introduction

The positive impact of physical activity on individual health and well-being is widely recognized in the available literature, including its contribution to lowering the risk of mortality [1]. Resistance training (RT) represents an exercise modality that is known to provide unequivocal health benefits if practiced at least twice per week [2]. In addition, RT has a fundamental role in the enhancement of performance and the mitigation of injury risk in sports [3, 4]. Therefore, investigating RT continues to be highly relevant in the field of sports science.

There is a significant underrepresentation of females in human research [5, 6]. This also holds true for research

conducted in the field of sports science and exercise physiology, with much of the existing evidence being derived from studies conducted primarily on male participants [7]. Unfortunately, such approach has led to a marked gap in understanding female-specific adaptations to RT [5]. Contemporary evidence highlights the physiological differences between males and females across most biological systems, including the neuromuscular system [8–10]. Also, there is undisputed evidence that such sex-based physiological and anatomical differences influence the limits of human motor performance [8–10]. Authors such as Roberts et al. highlight the importance of conducting well-designed studies focusing on comparisons between males and females, reasoning that sex differences in the physiology of RT remain poorly characterized [11]. This issue is particularly relevant because several organizations have increasingly advocated for sex equality in sports. As an example, the 2024 Summer Olympics in Paris featured an equal distribution of male and female athletes, for the first time in history [12].

Recent conclusions point toward a similar relative strength gain potential in both sexes, as well as an equivalent hypertrophic potential (in relative terms). These comparisons are generally valid for trained and untrained individuals together [6, 8, 11, 13–15]. Despite this, the published data often exhibit significant variability and heterogeneity, revealing the need for further research [11], possibly focusing on specific populations. Moreover, sex differences in maximal strength assessments are more pronounced during concentric (CON) rather than eccentric (ECC) or isometric (ISO) contractions [14]. These disparities concerning sex differences in maximal strength assessments seem to be mainly attributed to males' larger muscle mass [8, 14]. Differences between sexes are also observed in muscle fiber distribution [14–16]. While females have a higher proportion of type I muscle fibers relative to total muscle cross-sectional area (CSA), males have a greater percentage of type II fibers occupying their muscle mass [11, 13–15]. Females exhibit greater fatigue resistance than males, especially during ISO contractions performed at the same relative intensity [17, 18].

Focusing on psychological and motivational differences, RT tends to hold a higher priority among males in their hierarchy of preferred physical activities [14]. While males are more likely to favor RT over other activities, females generally show a preference for stretching, yoga and group aerobics [14]. When considering RT prescription, Refalo et al. recommended a similar approach for both sexes (untrained and resistance-trained individuals), tailoring the prescription to long-term goals (e.g., aesthetics or performance) and personal characteristics (e.g., enjoyment, perceptions of discomfort, preferences) [15]. On the contrary, other authors have claimed that it remains unclear whether individuals of different sexes should follow the same RT prescription [11].

While focusing on these dissimilarities and gaps, this brief narrative review sought to present detailed information about the physiological and performance-related sex differences in response to RT, among young and healthy adults. It aimed to describe sex-based differences in RT, including maximal strength, speed strength, strength endurance and muscle hypertrophy. Based on the analysis of recent literature, it intends to enhance the understanding of sex differences in response to RT, contributing to evidence-based insights that might help the optimization of training and performance outcomes for both sexes.

For this review, a semi-structured literature search was conducted across sport-science databases, including PubMed, Scopus, and Web of Science. Preference was given to research published in the last five years (2020–2025), whereas older studies were additionally identified by screening the references of more recent articles. Only peer-reviewed studies conducted in human adults and written in English were included. The search focused on RT and RT adaptations, and included the following terms: "sex differences", "gender differences", "resistance training", "strength training", "muscle performance", "neuromuscular adaptations", "maximal strength", "muscle size", "muscle hypertrophy", "speed strength", "strength endurance", "muscle fatigue", "time to task failure", "time course of adaptation", "muscle architecture", "muscle thickness", "cross-sectional area".

Sex differences in muscle strength and hypertrophy

Despite the lack of statistical significance, some studies have reported small sex-based disparities in grip strength from an early age (10 years old) [14, 19, 20]. A pronounced increase in grip strength occurs in males during adolescence, while females exhibit approximately 20% less grip strength than their male counterparts at this stage of development [14, 19, 20]. Both males and females reach their peak-strength levels between the age of 30 and 39 years [14, 19, 20].

In a review conducted by Nuzzo, female adults' strength was set around 40–55% of male strength in the bench- and chest-press one-repetition maximum (1RM) [14]. For the leg press and the back squat 1RM, female strength was approximately 60–65% of male strength [14]. Thus, sex differences in absolute strength seem to be more evident for the upper rather than the lower body [14]. Considering sex differences among powerlifters, higher strength-to-body-mass ratios (i.e., relative strength) have been observed in males compared to females, with approximate values of 1.5 vs. 1.0 for the bench press and 2.1 vs. 1.6 for the back squat, respectively [14, 21, 22]. In this case, differences were of similar magnitude when comparing sexes for both the upper

and the lower body. Refalo et al. also used this categorization to measure body regions and found that relative changes in muscle mass with training were similar between sexes, regardless of the analyzed body region [15]. Recent data examining the adaptations to RT in young to middle-aged adults (including both trained and untrained participants) [11] indicate that females may have a greater potential to enhance upper-body strength (relative terms), independently of training status, duration (weeks), frequency (days per week), and number of joints involved in exercise [11]. However, mixing untrained with trained individuals may have led to misleading conclusions [23]. For this reason, these findings should be interpreted with caution.

Muscle strength and performance

Sex differences in physical performance become more evident in tasks and events that rely primarily on muscle strength and power (favoring males) [8]. Maximal muscle strength is associated with muscle CSA [24, 25] and other factors, which are beyond the scope of this review (to read more about this topic, see Kuschel et al., for instance [26]). Power, which is determined by the interaction between force and velocity of contraction, is also generally greater in the skeletal muscles of males than in those of females, with velocity being influenced by fascicle length (FL) [8, 27, 28]. Females exhibit greater fatigue resistance than males during ISO strength endurance tests, but this difference becomes less obvious in response to dynamic contractions and varies depending on the analyzed muscle group [17, 29]. ISO and dynamic fatigue assessments show poor correlation, highlighting the complexity of assessing sex differences in such physiological conditions [17, 18, 29, 30].

Maximal strength

Hollander et al. studied 6 upper- and lower-body exercises in a small sample of young adults (10 participants of each sex, with ≥ 1 year of RT experience), assessing ECC and CON 1RMs [31]. The observed sex differences in muscle strength were generally smaller during ECC, compared to CON 1RM. Females exhibited higher ECC: CON strength ratios and, within this sex, ratios were greater in the upper rather than in the lower body. The most pronounced sex differences were observed in the military press and bench press, followed by the leg curl and leg press exercises.

More recently, a meta-analysis indicated that ECC muscle strength is approximately 40% superior to CON muscle strength, with the ECC:CON ratio being primarily influenced by movement velocity (under isokinetic conditions) and, to a lesser extent, by age or sex. Nonetheless, the ratio was reported to be slightly higher in females and older adults, and it is worth noting that this study did not account for the participants' training status [32]. The magnitude of sex differences in muscle strength is often greater in CON strength tests than in ISO or ECC tests [14, 31, 32]. Moreover, the size of the sex difference during ISO contractions varies with the muscle length at which the contraction is performed [14]. As a final illustration, a study comparing ISO squat strength and muscle activity in strength-matched males and females reported no significant sex differences. A critical consideration is that this study focused on a lower-limb ISO task performed by adults with prior RT experience [33]. All of these findings demonstrate that multiple factors can influence the outcomes of strength assessments.

Maximal strength is often assessed using 1RM (CON) and maximal voluntary ISO contractions (MVIC) [28, 30, 31]. Both methods are widely accepted, with MVICs often being favored because they offer the possibility of quantifying the force–time characteristics, are efficient, have high test–retest reliability, are sensitive to strength changes, involve minimal coordination requirements, display low risk of injury, and elicit lower levels of fatigue when compared to 1RM testing [34]. However, 1RM testing remains popular because it does not require the use of expensive equipment and is highly specific (practical standpoint), as it closely replicates exercise movements [34, 35].

Little to no evidence exists for sex differences in voluntary activation level [8, 14, 36]. Therefore, the sexual dimorphism in muscle maximal strength is most likely explained by differences in muscle mass rather than by neural-dependent mechanisms [8, 14, 37]. As previously outlined, most studies indicate that both males and females experience significant increases in maximal strength following RT [13, 38–40]. Therefore, the overall message emerging from all these data is that, while males typically exhibit greater absolute strength (and strength improvements) due to higher initial muscle mass and testosterone serum levels, relative strength improvements (percentage gains) are often comparable between sexes [8, 11, 14, 27].

Speed strength

Speed strength, or the ability to exert force rapidly, is crucial for athletic performance [27]. Since most sports require explosive movements, such as sprinting, jumping, and weightlifting, the ability to generate force rapidly is particularly relevant from a performance standpoint [41].

The rate of force development (RFD), which can be assessed during MVICs [42], does not seem to differ between sexes when normalized to maximum force [43]. Conversely, males tend to exhibit higher RFD in absolute terms, potentially due to sex differences in muscle fiber composition and overall muscle mass [43].

In athletes competing in strength and power-related events (e.g., weightlifting, rugby, track and field), with ≥ 3

years of RT experience, it was shown that the discrepancies in lower-body motor performance (strength and power — squat 1RM, deadlift 1RM and countermovement jump test) become dissipated after adjusting for lean body mass [27]. In contrast, males demonstrated higher maximal strength and power adjusted for lean body mass in the upper body (bench press 1RM and bench throw test). Still within this study, sex differences in power, when adjusted for muscle thickness, remained significant for both the upper and lower body. More precisely, females exhibited 61.2% less power in the upper body and 44.2% less power in the lower body. Finally, the FL of the assessed muscles was longer in males [27]. Greater muscle FL has been suggested to positively influence the muscle's maximum contraction velocity, thereby enhancing power output [27, 44, 45]. Males have greater skeletal-muscle volume and a relatively larger fast-contracting type II fiber area than females [8, 27]. Consequently, the normalized anaerobic power is significantly greater in males compared to females of similar training status and age, surpassing the sex differences observed in maximal strength (i.e., when strength is assessed independently of the time factor) [8, 27].

When examining soccer athletes, short-term power training (6 weeks of plyometric training; countermovement jump, drop jump, and medicine ball throw tests) has been suggested to induce similar adaptations in both sexes [46]. Other investigators have confirmed that a 6-week RT protocol, focusing on upper-body plyometric training, is highly effective for inducing adaptive responses in performance in physically active young adults[1] [28]. However, in this case, males exhibited greater adaptive responses (in relative terms) than females in the horizontal. medicine ball throw, peak power output, and mean power output (upper-body 30-s Wingate test) [28]. The underlying causes of these outcomes remain largely unclear, but factors such as muscle fiber type composition, muscle cross-sectional area [45], and glycolytic enzymatic activity were advanced as potential explanations [28]. The main conclusion of this study was that sex should be taken into account when designing upper-body power training programs [28].

Velocity-based training (VBT) is a method that emphasizes movement speed and may be a useful tool to enhance speed strength [47, 48]. The load-velocity relationship is strongly influenced by an individual's muscle strength levels and training [48, 49], with sex playing a role in shaping this relationship within each strength category (i.e., experienced or non-experienced), among young adults [49, 50]. Thus, extracting the individualized load-velocity profiles is recommended [48, 49]. For the lower body strength (Smith machine back squat), males exhibit a steeper load-velocity profile (LVP) compared to females, with higher velocity intercepts and faster mean concentric velocities across all relative loads, including the 1RM [49]. In terms of upper body strength (Smith machine bench press), LVP also varies between sexes: males tend to have a steeper profile and higher mean concentric velocities at lighter loads, while females often show faster mean velocities at heavier loads [51]. These results underscore the value of monitoring LVP throughout RT to uncover potential sex-specific adaptations.

There is preliminary evidence that both sexes can improve speed strength similarly with VBT, although it seems that females may require a greater velocity loss than males, particularly in the upper body, to maximize power and strength development over time [48, 50]. This conclusion was highlighted in the work of Rissanen et al., which examined the Smith machine bench press and back squat performance during VBT in participants with at least one year of experience in RT and in these specific exercises [50]. The researchers implemented 8-week training protocols using 20% and 40% velocity loss, with training loads of ~ 65–75% 1RM. Notably, while both female groups showed comparable improvements in the lower body performance (squat, countermovement jump, and vastus lateralis cross-sectional area), those included in the 40% velocity loss group appeared to show greater improvements in maximal strength and submaximal-load velocity in the bench press, with large effect sizes. Importantly, this pattern did not emerge in male participants. In accordance, to achieve optimal adaptation, power training programs for females may need a higher repetition volume than those typically used for males [50]. Given the limited scope of these data, such findings cannot be confidently generalized, highlighting the need for additional research in this domain.

Strength endurance

Strength endurance represents the ability to sustain repeated muscle contractions over time [52]. It is influenced by RT in both sexes [52, 53]. Jones et al. reported that training with low resistance and a high number of repetitions generally leads to greater improvements in muscular endurance [52]. Evidence generally shows that females are more fatigue-resistant than males, particularly during low-intensity, sustained, unilateral isometric contractions [17]. However, most studies focus on unilateral tasks, and there are limited data on sex-specific fatigue responses during bilateral contractions, which recruit greater muscle mass [30, 54]. At contraction intensities over 50% MVIC, blood-flow occlusion is thought to occur similarly in both sexes, reducing perfusion-related differences and minimizing disparities in

[1] The participants' training experience was not fully detailed; only familiarity with upper- and lower-body plyometric training was reported.

time-to-task failure [8, 17, 30]. This may explain why some studies report no sex differences at higher force outputs [54].

Considerably less is known about sex differences in the fatigability of dynamic contractions and the underlying mechanisms [8, 17, 29]. Current evidence indicates that males and females do not fatigue in the same manner during repeated slow-velocity contractions, although this disparity becomes smaller when the contractions are executed at higher speeds [29]. Notably, fatigability in dynamic tasks is highly dependent on the specific features of the task, including contraction velocity and the muscle group involved. Findings show that females generally fatigue less than males in the elbow flexors during slow, but not fast, dynamic contractions [29]. In those dynamic tasks where sex differences are clearly observed, the primary drivers appear to be contractile mechanisms within the muscle rather than central neural factors. While voluntary activation does decline following fatiguing dynamic exercise, this central component is affected to a similar degree in both sexes [29].

In a cross-sectional study, using unilateral biceps curls at 50% 1RM to failure, it was shown that females are able to complete more repetitions and exhibit less strength loss over time than males [18]. The paucity of studies focusing on strength endurance highlights the necessity for more targeted research. Despite the abundance of studies on fatigue assessments, to our knowledge, research comparing sex differences in chronic adaptation at the level of strength endurance remains quite limited at this stage.

Muscle hypertrophy

Muscle hypertrophy refers to the growth of skeletal-muscle mass resulting from the repeated exposure to resistance exercise [55, 56]. Despite the relevance of several physiological differences between sexes, both males and females experience a similar magnitude of relative increase in muscle size with training [8, 11, 14, 57]. Roberts et al. reported no significant sex differences in relative muscle hypertrophy, indicating that females are just as capable as males of achieving proportional increases in muscle size from their baseline. [11]. As a result, while absolute gains tend to slightly favor males, relative increases in hypertrophy appear to be of similar magnitude between sexes [15]. Previous research suggests that long-term experience with RT may modify the physiological responses to training [58] and influence muscle fiber type conversions, potentially impacting sex-specific hypertrophy outcomes [15, 59]. More recent studies denote that RT experience does not seem to influence these differences [15]. However, these findings warrant cautious interpretation, given that only a small number of studies included adequate information on participants' training experience, limiting the confidence in sex-comparison outcomes [15].

Skeletal muscle mass and architecture

The measurement of skeletal muscle mass using imaging techniques enables the quantitative assessment of muscular adaptations across various research settings, including RT [15, 60–62]. One commonly used method is the Brightness mode (B-mode) ultrasound, a cost-effective tool that requires less execution time compared to the gold-standard techniques for such assessments (i.e., magnetic resonance imaging and computed tomography) [61, 62]. The use of B-mode ultrasound for this particular purpose has been shown to be both valid and reliable, and its application in research has increased significantly in recent years [61–63]. Absolute differences between sexes in muscle mass become evident when using direct measures such as MT and muscle CSA [15]. Additionally, there is compelling evidence that males have more muscle mass not only in absolute terms, but also as a proportion of total body mass [14, 39, 64]. Finally, it should be noted that males also exhibit more muscle mass than females in the upper body as a proportion of total body mass [14, 27, 63].

Muscle cross-sectional area Sex differences in muscle mass are well-documented [63, 65], with males exhibiting greater absolute and relative muscle mass [63], as outlined above. In the upper body, the CSA of the female elbow flexors is approximately 50% of that in males [24], while their strength is about 55% of male values [14]. Hubal et al. reported that, at baseline, among young adults, females have 36% less CSA in the biceps brachii compared to males [39]. Despite these differences in CSA, strength per unit CSA (muscle quality) does not differ between sexes according to Miller et al. [24]. At the level of the lower body, Miller et al. reported that the knee-extensor CSA in females is approximately 75% of that typically seen in male values [24]. According to their report, the vastus lateralis CSA is larger in younger compared to older adults and in males compared to females. The rectus femoris CSA also shows the same trend in males compared to that observed in females [24].

Muscle thickness MT correlates with muscle CSA [66] and can be described as the vertical distance measured between the superficial and deep aponeurosis [67]. Consequently, this measurement is relevant for monitoring muscle adaptation induced by various stimuli, such as RT [60, 68]. In general, past reports show sex differences in MT with males having greater values than females across numerous muscle groups [27, 39, 40]. This difference is primarily attributed to higher baseline muscle mass and larger CSAs in males [66].

One study examining the time course of adaptation to RT reported no significant sex differences in the percentage change of MT following 12 weeks of progressive overload [40]. Nonetheless, baseline differences between sexes in MT

were observed both in the upper and the lower body [40]. Regardless of sex, this particular study also found a greater relative increase in MT in the upper (~ 10–12%) compared to the lower body (~ 7–9%) at the post-training time point. In another study, both males and females showed increases in MT after RT, with similar relative changes in response to a given training volume [68]. Despite this, in absolute terms, the hypertrophic response to RT tends to be consistently larger for males [40]. Finally, it should be noted that changes in MT along the muscle belly are not always uniform, and both sexes may exhibit regional variability in hypertrophy within a given muscle [67].

Muscle fascicle length Muscle FL is suggested to be a key determinant of a muscle's maximum contraction velocity [44, 45]. Longer fascicles consist of a greater number of sarcomeres in series, enabling a faster shortening velocity [27, 44]. One study conducted on a mixed sample of trained and untrained adults found significant sex differences in relative FL across the lower body muscles. Males exhibited longer fascicles in the vastus lateralis, whereas females had longer fascicles in the gastrocnemii [69]. These findings are further supported by Bartolomei et al. who also reported significantly longer FL in the vastus lateralis of males compared to that seen in females [27]. This is relevant because there is some evidence that greater FL is linked to superior sprint running performance in athletes of both sexes, whether due to genetic predisposition or training adaptations [70]. However, in another study, Abe et al. observed no sex differences in FL (vastus lateralis, gastrocnemius medialis and gastrocnemius lateralis) among elite sprinters [70]. Kubo et al. found longer absolute values of FLs in young males compared to young females across all analyzed muscles (vastus lateralis, medial gastrocnemius and triceps brachialis) [71]. Despite this, the relative FL (adjusted for limb length) of the vastus lateralis was significantly longer in females. No significant differences between sexes in relative FL were found for the triceps brachialis [71]. Stasinaki et al. studied a group of 9 young and novice female participants who completed a 6-week training regimen (two sessions per week), consisting of 6 sets of 6-RM unilateral cable elbow extensions [72]. Their results showed that, although MT (and strength) increased after training, this was not the case for FL [72].

Fiber type distribution There is strong evidence suggesting that females have higher type I/II fiber area ratios than males, while males display greater type II/I fiber ratios [16, 24]. This difference arises from males having larger CSAs for all muscle fiber types and a greater proportion of type II fibers [16]. In contrast, females show a higher distribution and area percentage of type I fibers [16]. Since type II fibers generate more force and power than Type I fibers (due to their size) [14, 16, 24, 59], the larger muscle CSA and higher type II fiber proportion in males provides a partial explanation as to why female muscle strength is ~ 60% of that typically seen in males [14, 16]. Conversely, the greater fatigue resistance observed in certain resistance exercises among females can be largely attributed to their greater proportion of type I fibers [8, 17, 29], which exhibit a higher oxidative capacity [14, 16, 24]. Refalo et al. provide evidence supporting the possibility of sex-specific differences in muscle fiber type hypertrophy [15]. Their meta-analysis showed that males are more likely to experience greater growth in type I muscle fibers, both in absolute and relative terms. In contrast, differences in type II muscle fiber hypertrophy between sexes were minimal, with females possibly showing a slight advantage in relative growth [15]. These outcomes may have been influenced by methodological variability across studies, such as the number of muscle fibers analyzed or the grouping of type IIa and type IIx fibers, prompting the authors to urge caution in interpreting the results [15]. Nonetheless, these findings suggest that males and females may adapt differently to RT at the muscle fiber level, particularly in type I fibers [15]. It is also important to note that not all female muscles exhibit a higher percentage of type I fibers [16]. For instance, the average type I fiber composition of the gastrocnemius medialis is 47.8% in females compared to 61.8% in males [16]. These variations underscore the importance of interpreting generalized data with caution and emphasizing the need for context-specific assessments that account for the particular muscle, population, and training background under investigation.

Time course of adaptation

Only a few studies have explored the time course of RT adaptations between sexes. Abe et al. examined previously untrained individuals and observed distinct sex-specific adaptation timelines, with males improving upper-body strength only by week 6 and lower-body strength by week 2, while females showed significant gains in both regions by week 4 [40]. Even though both sexes reached a similar magnitude of muscle hypertrophy at the end of this study, differences emerged in the time course of muscle growth. Specifically, while females increased the thickness of the biceps brachialis by week 8, this was seen in males immediately at week 4 ("Muscle thickness" Section). However, thessse findings have limited applicability to trained populations, and the authors emphasized sex similarities despite reporting timing differences. Subsequent research conducted by Kell et al. assessed trained individuals every four weeks and found that, although males remained stronger in absolute terms, females demonstrated greater relative strength gains [73]. Both sexes achieved substantial strength improvements after 12 weeks of periodized RT. However, slight

sex differences in upper-body strength may have been confounded by the lack of control for differences in prior training practice/experience.

More recently, Santos Junior et al. investigated active adults with prior experience in RT and found that, although relative strength gains were similar between sexes, the progression of power and muscular endurance differed (lower-body strength) [74]. Overall, females demonstrated greater improvements in dynamic muscular endurance, with most gains occurring in the later phase of the intervention, whereas males showed their largest improvements during the early phase. In contrast, power increased more in males across all testing points, with females showing their largest improvement from baseline at week 6. Despite this, it should be noted that the absence of individualized training loads may have influenced these outcomes. Taken together, although some research has examined sex differences in the time course of adaptation to RT, most studies remain focused on baseline-to-endpoint differences, rather than on the dynamic nature of change. Unfortunately, this leaves important gaps in our understanding of how, and to what extent, do males and females adapt differently to RT.

Additional considerations

In addition to the previously mentioned factors, the literature highlights other factors that might also interact with the adaptation to RT in both males and females, further complicating this issue. Although these aspects do not fall within the scope of this brief review, it is important to state their relevance on this matter. RT adaptations are influenced by factors such as training variables (e.g., volume, intensity, exercise, selection), individual characteristics (e.g., age, genetics), nutritional status, recovery practices, and psychological factors [14, 19, 21, 23, 32, 57, 63, 75, 76]. Consequently, sexual dimorphism in the adaptation to RT can only be inferred in circumstances involving a strict control of all these intervening factors.

Sex hormones are frequently associated with adaptations to RT [9, 77, 78]. However, some studies support the contention that factors other than sex-specific hormonal balance may contribute to muscle hypertrophy more profoundly [11, 15]. This notion is supported by data providing no evidence of sex differences in the relative improvements in muscle strength and size post-training [11, 15]. Another argument against this perspective is that estradiol also has the potential to promote muscle hypertrophy due to its anabolic properties [78, 79]. Additionally, acute elevations in anabolic hormones do not stimulate muscle protein synthesis post-exercise (which plays a key role in triggering muscle hypertrophy in response to repeated exposure to RT) [80]. Finally, there are some data supporting the lack of sex differences in post-exercise muscle protein synthesis [81]. Most of the available research did not control for the influence of the menstrual cycle subsequent to RT, and this precludes more definite conclusions on the interaction between sex hormones and chronic adaptation [15, 77, 80]. As importantly, further research is warranted to unravel if the time course of adaptation to RT follows a true sexually dimorphic pattern (at the level of gains in muscle strength and size). This is crucial because such information might enable fine-tuning the periodization of training to reach a targeted adaptation more effectively over time in both sexes.

Conclusion

Despite the growing interest in characterizing sex differences following RT, key knowledge gaps persist. Although there is enough evidence to suggest that, when completing the same training regimen, both sexes show a similar magnitude of relative increase in muscle strength and size over time, numerous other factors (e.g., muscle fiber composition, fatigue resistance, training preferences, fatigue) can still explain some of the observed dissimilarities in motor performance. Notably, females often display greater fatigue resistance, whereas males exhibit higher absolute strength and power due to anatomical and neuromuscular differences. One critical aspect requiring further research is the time course of adaptations to RT, particularly in females. Understanding the rate at which both sexes manifest strength, hypertrophy, and performance increments (training effect) as well as decrements (detraining effect) is of utmost importance. Ultimately, this would enable a more effective optimization of training periodization for both males and females. Longitudinal intervention studies, including intermediate assessments, might provide valuable insights into this fundamental question: do males and females adapt at the same rate in response to RT or is it necessary to implement sex-specific programming adjustments to improve the final outcomes of training?

Limitations

It is also important to acknowledge the limitations of this brief narrative review. Unlike systematic reviews that follow a strict methodological framework, this work can only serve as a general overview of recent findings in the field. While it synthesizes key aspects of the literature regarding sex differences in thsse adaptation to RT, its narrative structure implies that not all aspects were comprehensively analyzed. Future research should aim to refine and expand upon these discussions through systematic approaches. Nonetheless, this review contributes to the ongoing discussion on sex differences in response to RT and emphasizes the need for

more inclusive, well-designed studies that address current knowledge gaps.

Practical Applications

- Exercise prescription should account for sex-specific factors, training experience, exercise familiarization, type of muscle contraction, and the muscle groups involved, as these variables might influence performance outcomes and adaptations
- Coaches and practitioners should be aware that females may require higher training volumes (e.g., achieved through greater velocity loss thresholds) to optimize upper-body strength and power gains, particularly in experienced trainees ("Speed strength" Section)
- Lower-body exercises (such as the deadlift or the kettlebell swing) may show distinct adaptation patterns between sexes, especially at lower intensities (%RM) over training periods of ~ 3 months. In this sense, training progression may need to differ between males and females, as emerging evidence indicates that females may adapt at a different pace in strength-related qualities such as power and muscular endurance ("Time course of adaptation" Section)
- Given the limited research, practitioners should apply these insights cautiously and adapt training plans to individual responses rather than relying solely on sex-based assumptions

Author contributions MC and GVM conceptualized the study. MC conducted the literature search and analyzed the data. MC drafted the document and GVM critically revised the work.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript. Goncalo V. Mendonca serves this journal as section editor and he has not taken part to the handling and peer-review process for this article.

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References

1. Shailendra P, Baldock KL, Li LSK et al (2022) Resistance training and mortality risk: a systematic review and meta-analysis. Am J Prev Med 63:277–285. https://doi.org/10.1016/j.amepre.2022.03.020
2. World Health Organization (2020) WHO guidelines on physical activity and sedentary behaviour. World Health Organization, Geneva
3. Saeterbakken AH, Stien N, Pedersen H et al (2024) The connection between resistance training climbing performance, and injury prevention. Sports Med Open. https://doi.org/10.1186/s40798-024-00677-w
4. Dudley C, Johnston R, Jones B et al (2023) Methods of monitoring internal and external loads and their relationships with physical qualities, injury, or illness in adolescent athletes: a systematic review and best-evidence synthesis. Sports Med 53:1559–1593. https://doi.org/10.1007/s40279-023-01844-x
5. Elliott-Sale KJ, Minahan CL, de Jonge XAKJ et al (2021) Methodological considerations for studies in sport and exercise science with women as participants: a working guide for standards of practice for research on women. Sports Med 51:843–861. https://doi.org/10.1007/s40279-021-01435-8
6. Vanden Noven ML, Anselmo M, Tahsin CT et al (2023) A review of the historical use of sex as a biological variable in the American Journal of Physiology-Heart and Circulatory Physiology. Am J Physiol Heart Circ Physiol 325:H768–H773
7. James JJ, Klevenow EA, Atkinson MA et al (2023) Underrepresentation of women in exercise science and physiology research is associated with authorship gender. J Appl Physiol 135:932–942. https://doi.org/10.1152/JAPPLPHYSIOL.00377.2023
8. Hunter SK, Senefeld JW, Hunter SK et al (2024) Sex differences in human performance. J Physiol 602:4129–4156
9. Alexander SE, Pollock AC, Lamon S (2022) The effect of sex hormones on skeletal muscle adaptation in females. Eur J Sport Sci 22:1035–1045. https://doi.org/10.1080/17461391.2021.1921854
10. Joyner MJ, Hunter SK, Senefeld JW (2025) Evidence on sex differences in sports performance. J Appl Physiol 138:274–281
11. Roberts BM, Nuckols G, Krieger JW (2020) Sex differences in resistance training: a systematic review and meta-analysis. J Strength Cond Res 0:1–13
12. International Olympic Committee (2024) Factsheet: women in the Olympic movement. https://stillmed.olympics.com/media/Documents/Olympic-Movement/Factsheets/Women-in-the-Olympic-Movement.pdf. Accessed 7 Feb 2025
13. Lopez P, Radaelli R, Taaffe DR et al (2021) Resistance training load effects on muscle hypertrophy and strength gain: systematic review and network meta-analysis. Med Sci Sports Exerc 53:1206–1216
14. Nuzzo JL (2023) Narrative review of sex differences in muscle strength, endurance, activation, size, fiber type, and strength training participation rates, preferences, motivations, injuries, and neuromuscular adaptations. J Strength Cond Res 37:494–536
15. Refalo MC, Nuckols G, Galpin AJ et al (2025) Sex differences in absolute and relative changes in muscle size following resistance training in healthy adults: a systematic review with Bayesian meta-analysis. PeerJ 13:e19042. https://doi.org/10.7717/peerj.19042
16. Nuzzo JL (2024) Sex differences in skeletal muscle fiber types: a meta-analysis. Clin Anat 37:81–91. https://doi.org/10.1002/ca.24091

17. Hunter SK (2014) Sex differences in human fatigability: mechanisms and insight to physiological responses. Acta Physiol 210:768–789. https://doi.org/10.1111/apha.12234
18. Voskuil CC, Dudar MD, Carr JC (2024) Sex differences in fatigability during single-joint resistance exercise in a resistance-trained population. Eur J Appl Physiol 124:2261–2271. https://doi.org/10.1007/s00421-024-05445-y
19. Wang YC, Bohannon RW, Li X et al (2019) Summary of grip strength measurements obtained in the 2011-2012 and 2013-2014 National Health and Nutrition Examination Surveys. J Hand Ther 32:489–496. https://doi.org/10.1016/j.jht.2018.03.002
20. Perna FM, Coa K, Troiano RP et al (2016) Muscular grip strength estimates of the U.S. population from the national health and nutrition examination survey 2011-2012. J Strength Cond Res 30:867–874. https://doi.org/10.1519/JSC.0000000000001104
21. Latella C, Teo W-P, Spathis J, Van Den Hoek D (2020) Long-term strength adaptation: a 15-year analysis of powerlifting athletes. J Strength Cond Res 34:2412–2418
22. Ball R, Weidman D (2018) Analysis of USA powerlifting federation data from January 1, 2012–June 11, 2016. J Strength Cond Res 32:1843–1851
23. McKay AKA, Stellingwerff T, Smith ES et al (2022) Defining training and performance caliber: a participant classification framework. Int J Sports Physiol Perform 17:317–331. https://doi.org/10.1123/ijspp.2021-0451
24. Miller AEJ, Macdougall JD, Tarnopolsky MA, Sale DG (1993) Gender differences in strength and muscle fiber characteristics. Eur J Appl Physiol Occup Physiol 66:254–262
25. Narici M (1999) Human skeletal muscle architecture studied in vivo by non-invasive imaging techniques: functional significance and applications. J Electromyogr Kinesiol 9:97–103
26. Kuschel LB, Sonnenburg D, Engel T (2022) Factors of muscle quality and determinants of muscle strength: a systematic literature review. Healthcare. https://doi.org/10.3390/healthcare10101937
27. Bartolomei S, Grillone G, Di Michele R, Cortesi M (2021) A comparison between male and female athletes in relative strength and power performances. J Funct Morphol Kinesiol. https://doi.org/10.3390/jfmk6010017
28. Lu G, Duan Y (2024) Sex differences in the adaptations in maximal strength and anaerobic power to upper body plyometric training. Sci Rep. https://doi.org/10.1038/s41598-024-72234-0
29. Hunter SK (2016) Sex differences in fatigability of dynamic contractions. Exp Physiol 101:250–255. https://doi.org/10.1113/EP085370
30. Benitez B, Kwak M, Succi PJ et al (2024) Examination of sex differences in fatigability and neuromuscular responses during continuous, maximal, isometric leg extension. Physiol Meas. https://doi.org/10.1088/1361-6579/ad7fcd
31. Hollander DB, Kraemer RR, Kilpatrick M, Reeves GV (2007) Maximal eccentric and concentric strength discrepancies between young men and women for dynamic resistance exercise. J Strength Cond Res 21:34–40. https://doi.org/10.1519/R-18725.1
32. Nuzzo JL, Pinto MD, Nosaka K, Steele J (2023) The eccentric: concentric strength ratio of human skeletal muscle in vivo: meta-analysis of the influences of sex, age, joint action, and velocity. Sports Med 53:1125–1136. https://doi.org/10.1007/s40279-023-01851-y
33. Nimphius S, McBride JM, Rice PE et al (2019) Comparison of quadriceps and hamstring muscle activity during an isometric squat between strength-matched men and women. J Sports Sci Med 18:101–108
34. Warneke K, Wagner CM, Keiner M et al (2023) Maximal strength measurement: a critical evaluation of common methods—a narrative review. Front Sports Act Living. https://doi.org/10.3389/fspor.2023.1105201
35. Schoenfeld BJ, Grgic J, Van Every DW, Plotkin DL (2021) Loading recommendations for muscle strength, hypertrophy, and local endurance: a re-examination of the repetition continuum. Sports. https://doi.org/10.3390/sports9020032
36. Molenaar JP, McNeil CJ, Bredius MS, Gandevia SC (2013) Effects of aging and sex on voluntary activation and peak relaxation rate of human elbow flexors studied with motor cortical stimulation. Age (Omaha) 35:1327–1337. https://doi.org/10.1007/s11357-012-9435-5
37. Bishop P, Cureton K, Collins M (1987) Sex difference in muscular strength in equally-trained men and women. Ergonomics 30:675–687. https://doi.org/10.1080/00140138708969760
38. Kraemer WJ, Mazzetti SA, Nindl BC et al (2001) Effect of resistance training on women's strength/power and occupational performances. Med Sci Sports Exerc 33:1011–1025
39. Hubal M, Gordish-Dressman H, Thompson P et al (2005) Variability in muscle size and strength gain after unilateral resistance training. Med Sci Sports Exerc 37:964–972
40. Abe T, Dehoyos DV, Pollock ML, Garzarella L (2000) Time course for strength and muscle thickness changes following upper and lower body resistance training in men and women. Eur J Appl Physiol 81:174–180
41. van der Kruk E, van der Helm FCT, Veeger HEJ, Schwab AL (2018) Power in sports: a literature review on the application, assumptions, and terminology of mechanical power in sport research. J Biomech 79:1–14. https://doi.org/10.1016/j.jbiomech.2018.08.031
42. Maffiuletti NA, Aagaard P, Blazevich AJ et al (2016) Rate of force development: physiological and methodological considerations. Eur J Appl Physiol 116:1091–1116. https://doi.org/10.1007/s00421-016-3346-6
43. Hannah R, Minshull C, Buckthorpe MW, Folland JP (2012) Explosive neuromuscular performance of males versus females. Exp Physiol 97:618–629. https://doi.org/10.1113/expphysiol.2011.063420
44. Blazevich AJ, Giorgi A (2001) Effect of testosterone administration and weight training on muscle architecture. Med Sci Sports Exerc 33:1688–1693
45. Kumagai K, Abe T, Brechue WF et al (2000) Sprint performance is related to muscle fascicle length in male 100-m sprinters. J Appl Physiol 88:811–816
46. Ramírez-Campillo R, Vergara-Pedreros M, Henríquez-Olguín C et al (2016) Effects of plyometric training on maximal-intensity exercise and endurance in male and female soccer players. J Sports Sci 34:687–693. https://doi.org/10.1080/02640414.2015.1068439
47. Kirby TJ, Erickson T, Mcbride JM (2010) Model for progression of strength, power, and speed training. Strength Cond J 32:86–90
48. Weakley J, Mann B, Banyard H et al (2021) Velocity-based training: from theory to application. Strength Cond J 43:31–49
49. Fitas A, Santos P, Gomes M et al (2024) Influence of sex and strength differences on the load–velocity relationship of the Smith-machine back squat. Sport Sci Health 20:87–96. https://doi.org/10.1007/s11332-023-01069-5
50. Rissanen J, Walker S, Pareja-Blanco F, Häkkinen K (2022) Velocity-based resistance training: do women need greater velocity loss to maximize adaptations? Eur J Appl Physiol 122:1269–1280. https://doi.org/10.1007/s00421-022-04925-3
51. Torrejón A, Balsalobre-Fernández C, Haff GG, García-Ramos A (2019) The load-velocity profile differs more between men and women than between individuals with different strength levels. Sports Biomech 18:245–255. https://doi.org/10.1080/14763141.2018.1433872
52. Jones W, Coulter S (1994) Strength-endurance effects from three RT protocols with women. J Strength Cond Res 8:231–234

53. Stragier S, Baudry S, Carpentier A, Duchateau J (2019) Efficacy of a new strength training design: the 3/7 method. Eur J Appl Physiol 119:1093–1104. https://doi.org/10.1007/s00421-019-04099-5
54. Benitez B, Kwak M, Succi PJ et al (2024) No sex differences in time-to-task failure and neuromuscular patterns of response during submaximal, bilateral, isometric leg extensions. Eur J Appl Physiol 124:2993–3004. https://doi.org/10.1007/s00421-024-05508-0
55. Schiaffino S, Reggiani C, Akimoto T, Blaauw B (2021) Molecular mechanisms of skeletal muscle hypertrophy. J Neuromuscul Dis 8:169–183. https://doi.org/10.3233/JND-200568
56. Hirono T, Ikezoe T, Taniguchi M et al (2020) Relationship between muscle swelling and hypertrophy induced by resistance training. J Strength Cond Res 00:1–6
57. Stock MS, Oranchuk DJ, Burton AM, Phan DC (2020) Age, sex, and region-specific differences in skeletal muscle size and quality. Appl Physiol Nutr Metab 45:1253–1260
58. Damas F, Phillips S, Vechin FC, Ugrinowitsch C (2015) A review of resistance training-induced changes in skeletal muscle protein synthesis and their contribution to hypertrophy. Sports Med 45:801–807. https://doi.org/10.1007/s40279-015-0320-0
59. Plotkin DL, Roberts MD, Haun CT, Schoenfeld BJ (2021) Muscle fiber type transitions with exercise training: shifting perspectives. Sports. https://doi.org/10.3390/SPORTS9090127
60. Arruda BR, Soares ALC, Carvalho RF, Gomes PSC (2022) Reliability and measurement error of the proximal, medial and distal portions of the vastus lateralis muscle thickness measured with extended field of view ultrasonography. Kinesiology 54:107–114. https://doi.org/10.26582/K.54.1.11
61. Hernández-Belmonte A, Martínez-Cava A, Pallarés JG (2022) Pectoralis cross-sectional area can be accurately measured using panoramic ultrasound: a validity and repeatability study. Ultrasound Med Biol 48:460–468. https://doi.org/10.1016/j.ultrasmedbio.2021.10.017
62. Naruse M, Trappe S, Trappe TA (2022) Human skeletal muscle size with ultrasound imaging: a comprehensive review. J Appl Physiol 132:1267–1279. https://doi.org/10.1152/japplphysiol.00041.2022
63. Janssen I, Heymsfield SB, Wang Z, Ross R (2000) Skeletal muscle mass and distribution in 468 men and women aged 18–88 yr. J Appl Physiol 89:81–88
64. Behan FP, Maden-Wilkinson TM, Pain MTG, Folland JP (2018) Sex differences in muscle morphology of the knee flexors and knee extensors. PLoS ONE. https://doi.org/10.1371/journal.pone.0190903
65. Abe T, Kearns CF, Fukunaga T (2003) Sex differences in whole body skeletal muscle mass measured by magnetic resonance imaging and its distribution in young Japanese adults. Br J Sports Med 37:436–440
66. Franchi MV, Longo S, Mallinson J et al (2018) Muscle thickness correlates to muscle cross-sectional area in the assessment of strength training-induced hypertrophy. Scand J Med Sci Sports 28:846–853. https://doi.org/10.1111/sms.12961
67. Hodson-Tole EF, Lai AKM (2019) Ultrasound-derived changes in thickness of human ankle plantar flexor muscles during walking and running are not homogeneous along the muscle mid-belly region. Sci Rep. https://doi.org/10.1038/s41598-019-51510-4
68. Schoenfeld BJ, Contreras B, Krieger J et al (2019) Resistance training volume enhances muscle hypertrophy but not strength in trained men. Med Sci Sports Exerc 51:94–103. https://doi.org/10.1249/MSS.0000000000001764
69. Alegre LM, Lara AJ, Elvira LL, Aguado X (2009) Muscle morphology and jump performance: gender and intermuscular variability. J Sports Med Phys Fitness 49:320–326
70. Abe T, Fukashiro S, Harada Y, Kawamoto K (2001) Relationship between sprint performance and muscle fascicle length in female sprinters. J Physiol Anthropol Appl Human Sci. https://doi.org/10.2114/jpa.20.141
71. Kubo K, Kanehisa H, Azuma K et al (2003) Muscle architectural characteristics in young and elderly men and women. Int J Sports Med 24:125–130
72. Stasinaki AN, Zaras N, Methenitis S et al (2018) Triceps brachii muscle strength and architectural adaptations with resistance training exercises at short or long fascicle length. J Funct Morphol Kinesiol. https://doi.org/10.3390/jfmk3020028
73. Kell RT (2011) The influence of periodized resistance training on strength changes in men and women. J Strength Cond Res 25:735–744
74. Santos Junior ERT, De Salles BF, Dias I et al (2024) Sex differences in neuromuscular adaptations following 12 weeks of kettlebell swing training. Cureus 16:e70551. https://doi.org/10.7759/cureus.70551
75. Wohlgemuth KJ, Arieta LR, Brewer GJ et al (2021) Sex differences and considerations for female specific nutritional strategies: a narrative review. J Int Soc Sports Nutr. https://doi.org/10.1186/s12970-021-00422-8
76. Cholewa J, Trexler E, Lima-Soares F et al (2019) Effects of dietary sports supplements on metabolite accumulation, vasodilation and cellular swelling in relation to muscle hypertrophy: a focus on "secondary" physiological determinants. Nutrition 60:241–251. https://doi.org/10.1016/j.nut.2018.10.011
77. D'Souza AC, Wageh M, Williams JS et al (2023) Menstrual cycle hormones and oral contraceptives: a multimethod systems physiology-based review of their impact on key aspects of female physiology. J Appl Physiol 135:1284–1299. https://doi.org/10.1152/japplphysiol.00346.2023
78. Hansen M, Kjaer M (2014) Influence of sex and estrogen on musculotendinous protein turnover at rest and after exercise. Exerc Sport Sci Rev 42:183
79. Wackerhage H, Schoenfeld BJ, Hamilton DL et al (2019) Stimuli and sensors that initiate skeletal muscle hypertrophy following resistance exercise. J Appl Physiol 126:30–43. https://doi.org/10.1152/japplphysiol.00685.2018
80. Van Every DW, D'Souza AC, Phillips SM (2024) Hormones, hypertrophy, and hype: an evidence-guided primer on endogenous endocrine influences on exercise-induced muscle hypertrophy. Exerc Sport Sci Rev 52:117–125. https://doi.org/10.1249/JES.0000000000000346
81. West DWD, Burd NA, Churchward-Venne TA et al (2012) Sex-based comparisons of myofibrillar protein synthesis after resistance exercise in the fed state. J Appl Physiol 112:1805–1813. https://doi.org/10.1152/japplphysiol.00170.2012

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